FACTORY — Management System

One flow from customer order to delivery and payment.

ORDER → MATERIALS → PURCHASE → PRODUCTION → INVOICE → PAYMENT

System overview · July 2026 · Kampala, Uganda

What the system does

Factory runs the full operation of the plant — straws, sticks, aluminium foil, cling film, plates and lids — in one place. It replaces the spreadsheet workbook with a live system that every user shares: 37 finished products, 51 raw materials and the full bill of materials are already loaded, and every quantity on screen comes from an auditable stock ledger.

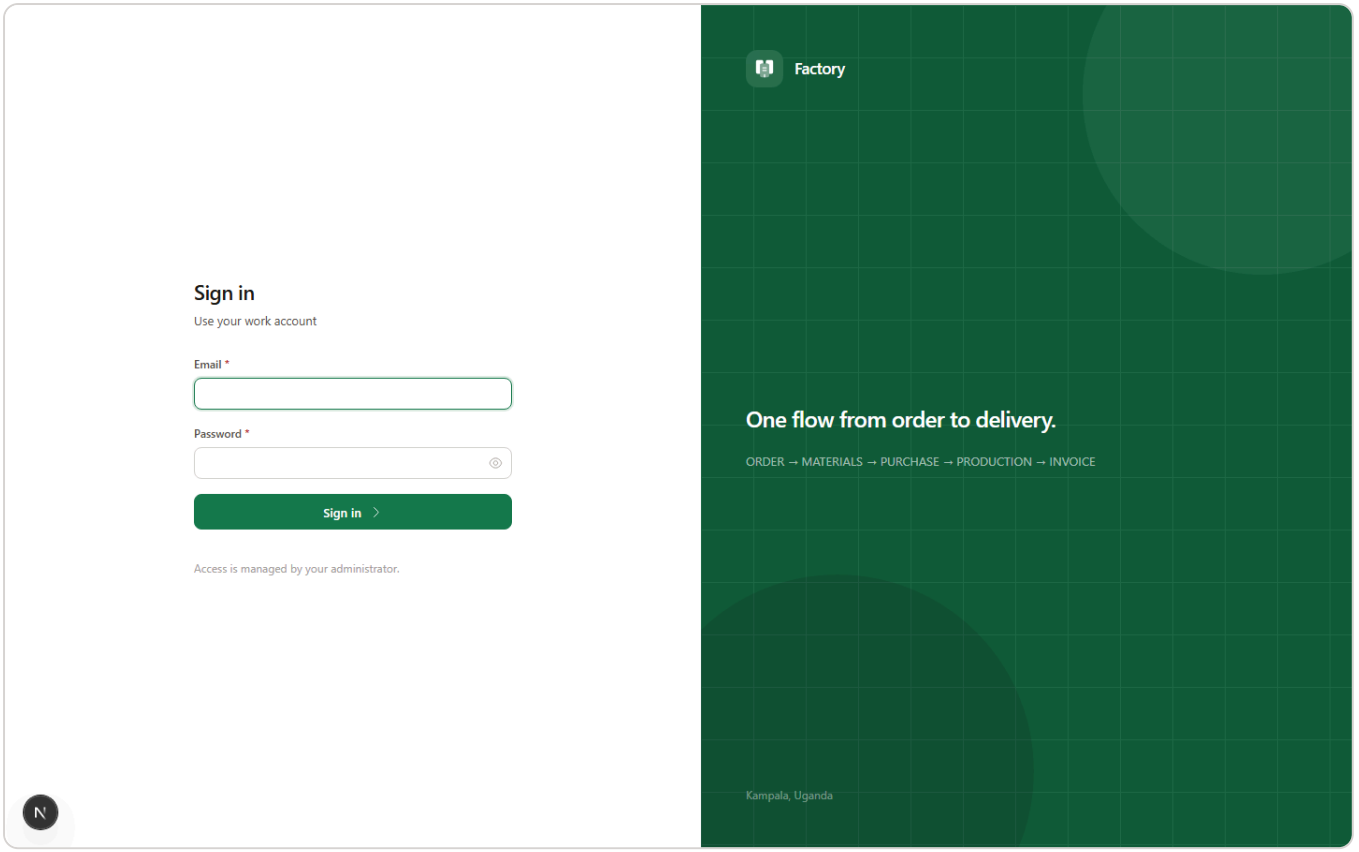
The system is built around the way an order actually moves through the factory:

- **Order** — a customer order is captured with products, quantities and rates.
- **Materials** — the BOM is exploded automatically and checked against free stock.
- **Purchase** — any shortfall becomes a purchase request, approved, ordered and received.
- **Production** — runs record what was produced and what was actually consumed.
- **Invoice & payment** — finished goods are invoiced and payments tracked to zero balance.

Every step is permission-controlled, audited, and updates all connected screens in real time. All amounts are in Uganda Shillings.

Signing in

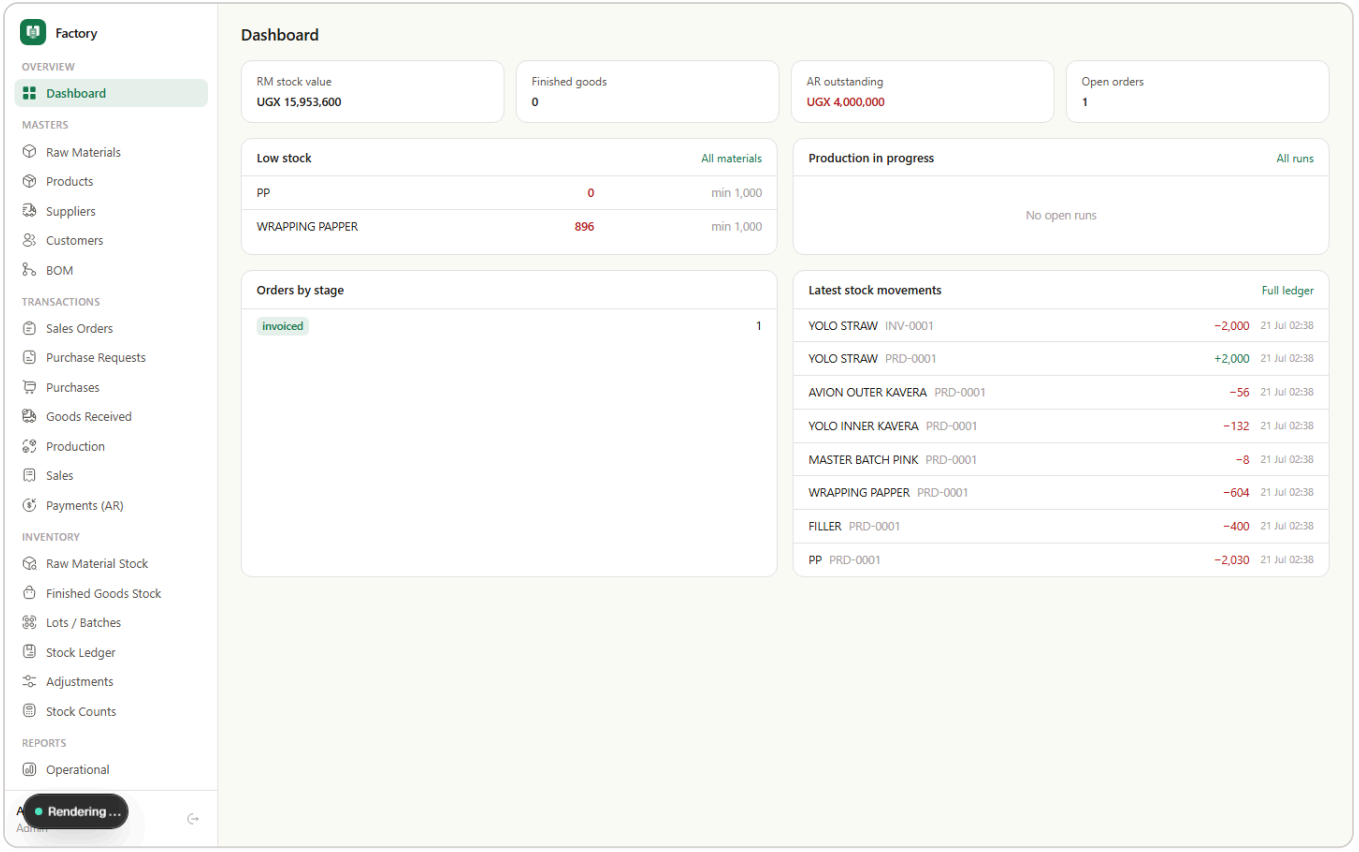
Each staff member has their own account with a role — Admin, Manager, Storekeeper, Production or Sales. What a person can see and do is set per role from the admin area.



Sign-in screen

Dashboard

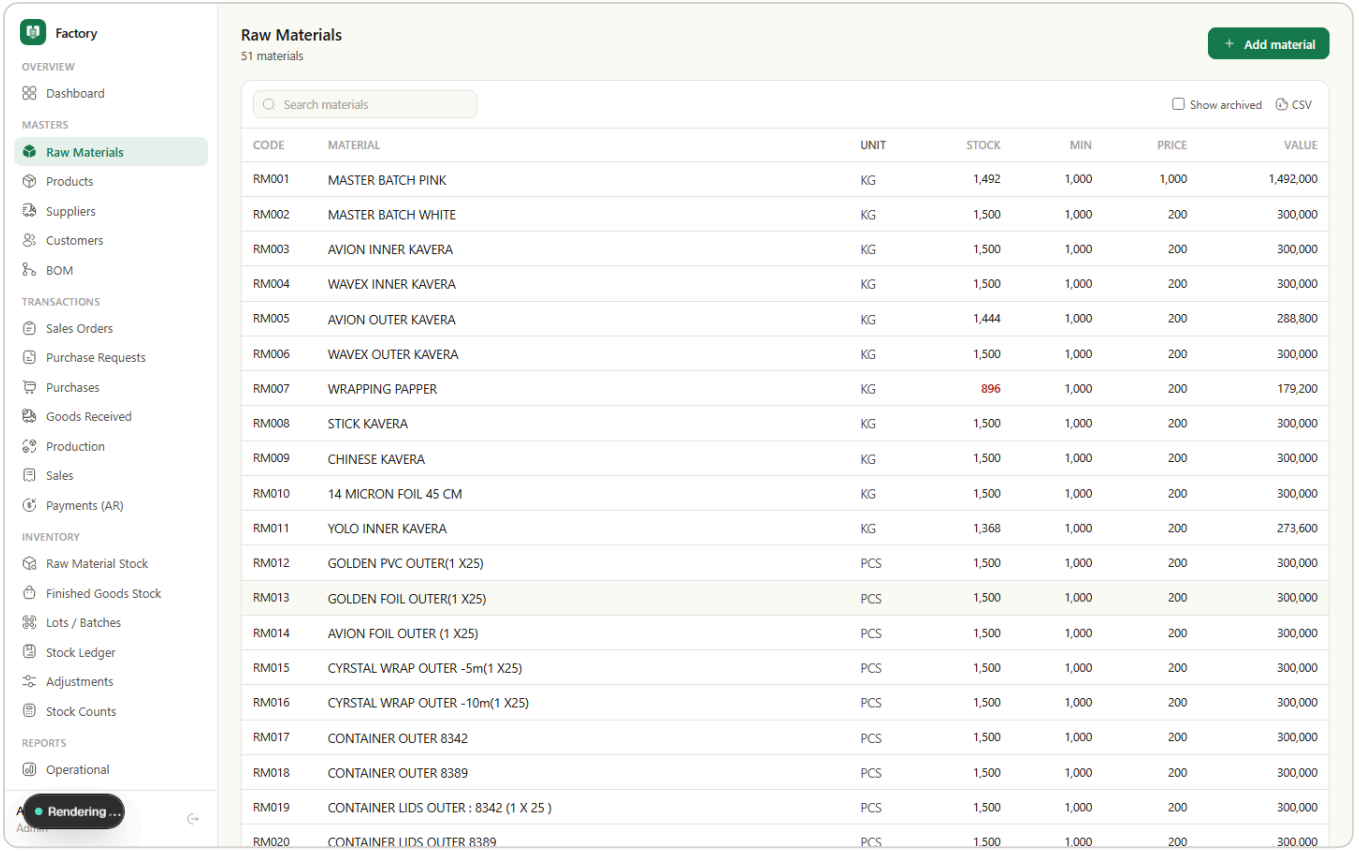
The dashboard is the morning view: raw-material stock value, finished goods on hand, outstanding receivables and open orders, plus materials running below their minimum, runs still in progress, and the latest stock movements. Items awaiting approval are one click away.



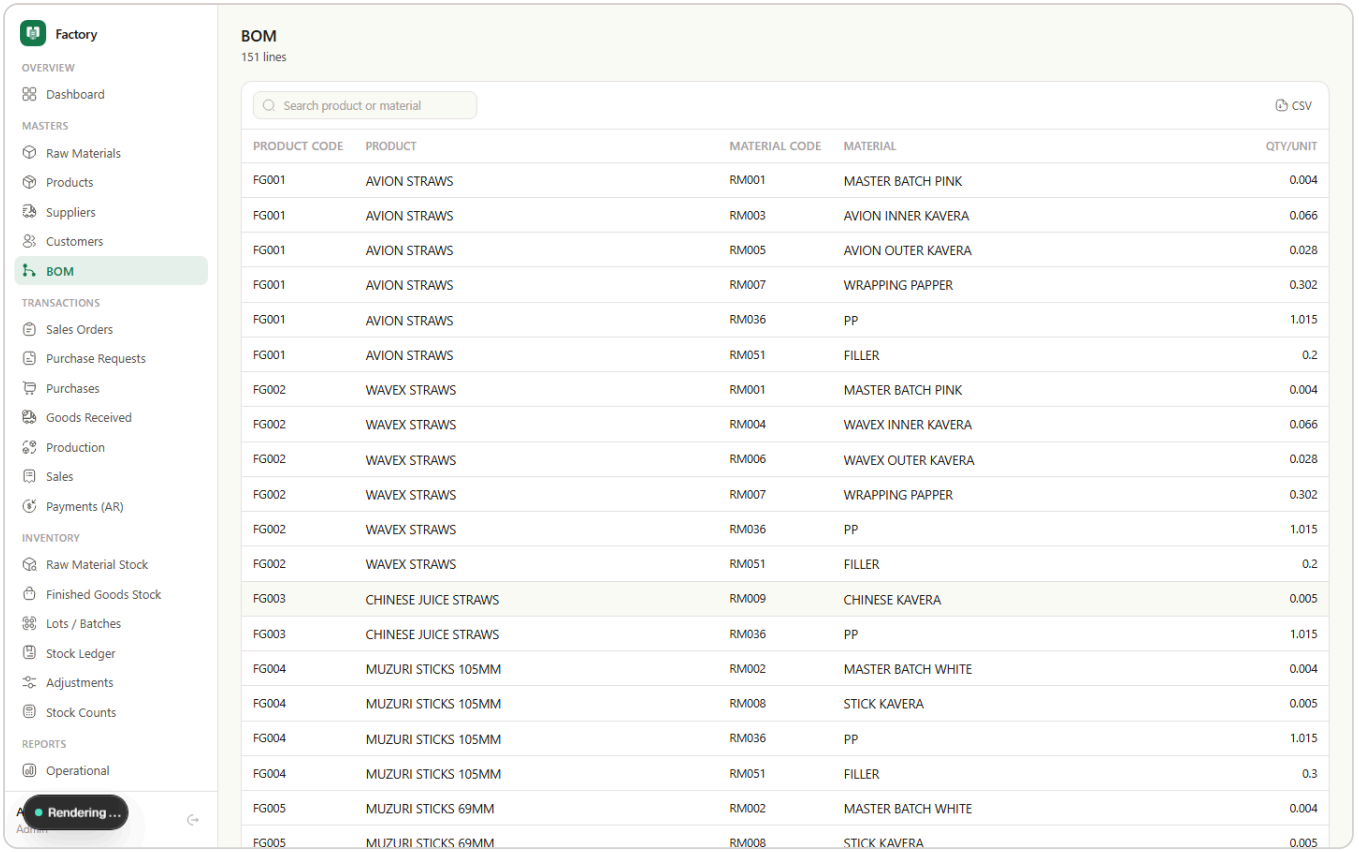
Live overview after a day's activity

Masters — materials, products, BOM

Raw materials carry a unit, price and minimum stock level; products carry their selling price and their bill of materials — the exact quantity of each material needed per unit produced. The BOM drives the material check and the production variance report, and can be edited line by line as recipes change.



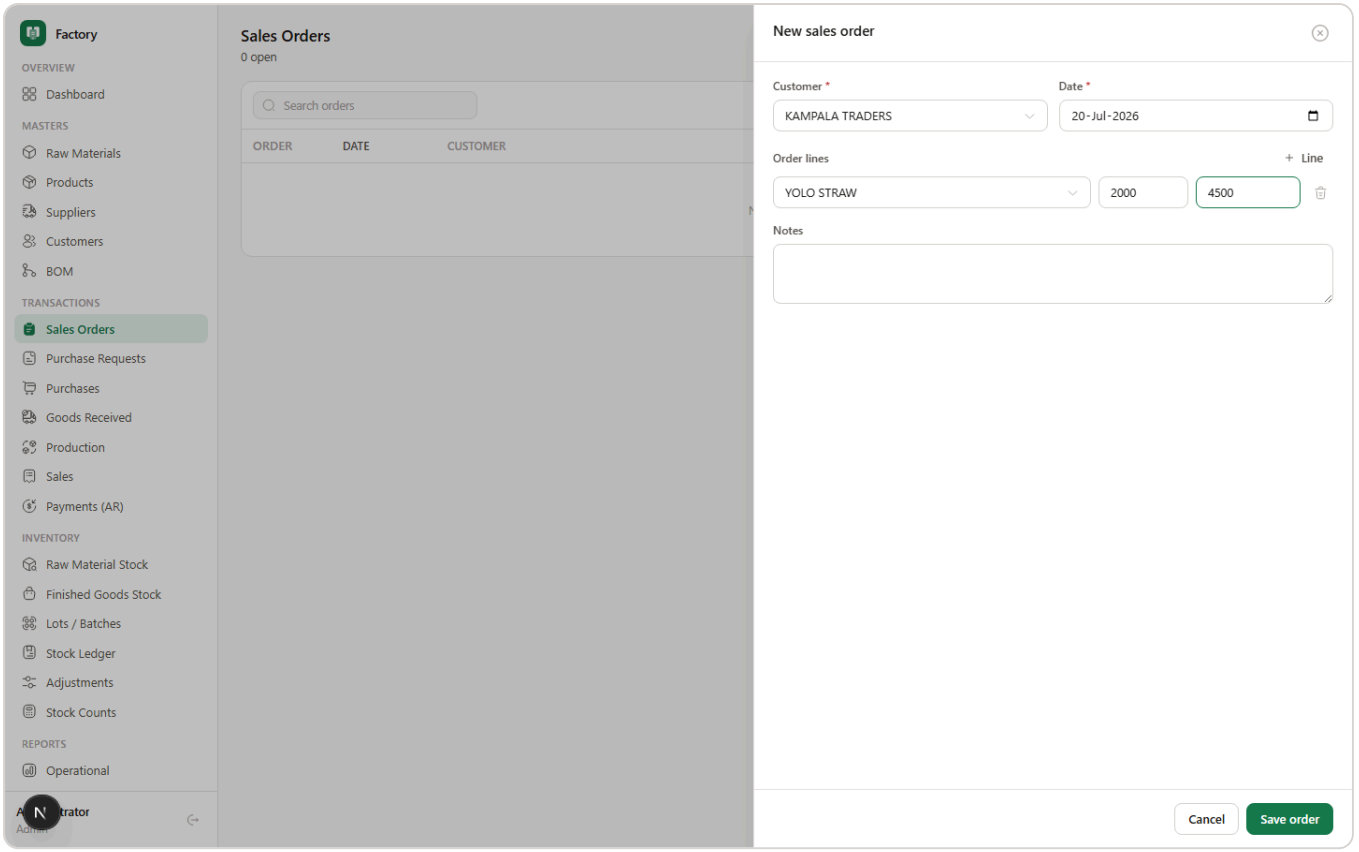
Raw materials with stock levels — low stock highlighted



Bill of materials across all products

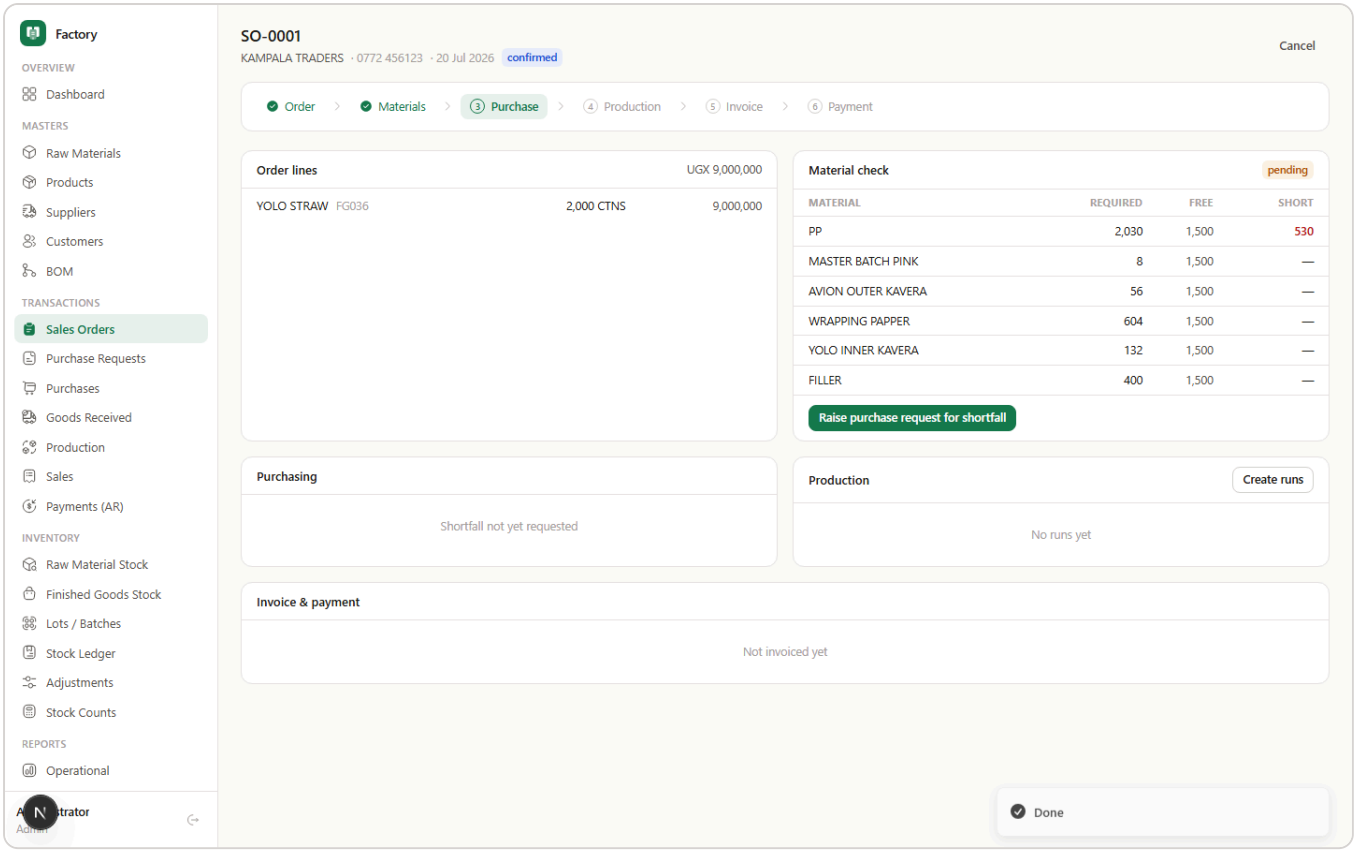
Taking an order

An order is captured in a slide-in form: customer, date, product lines and rates. Customers and suppliers are managed the same way.



New sales order

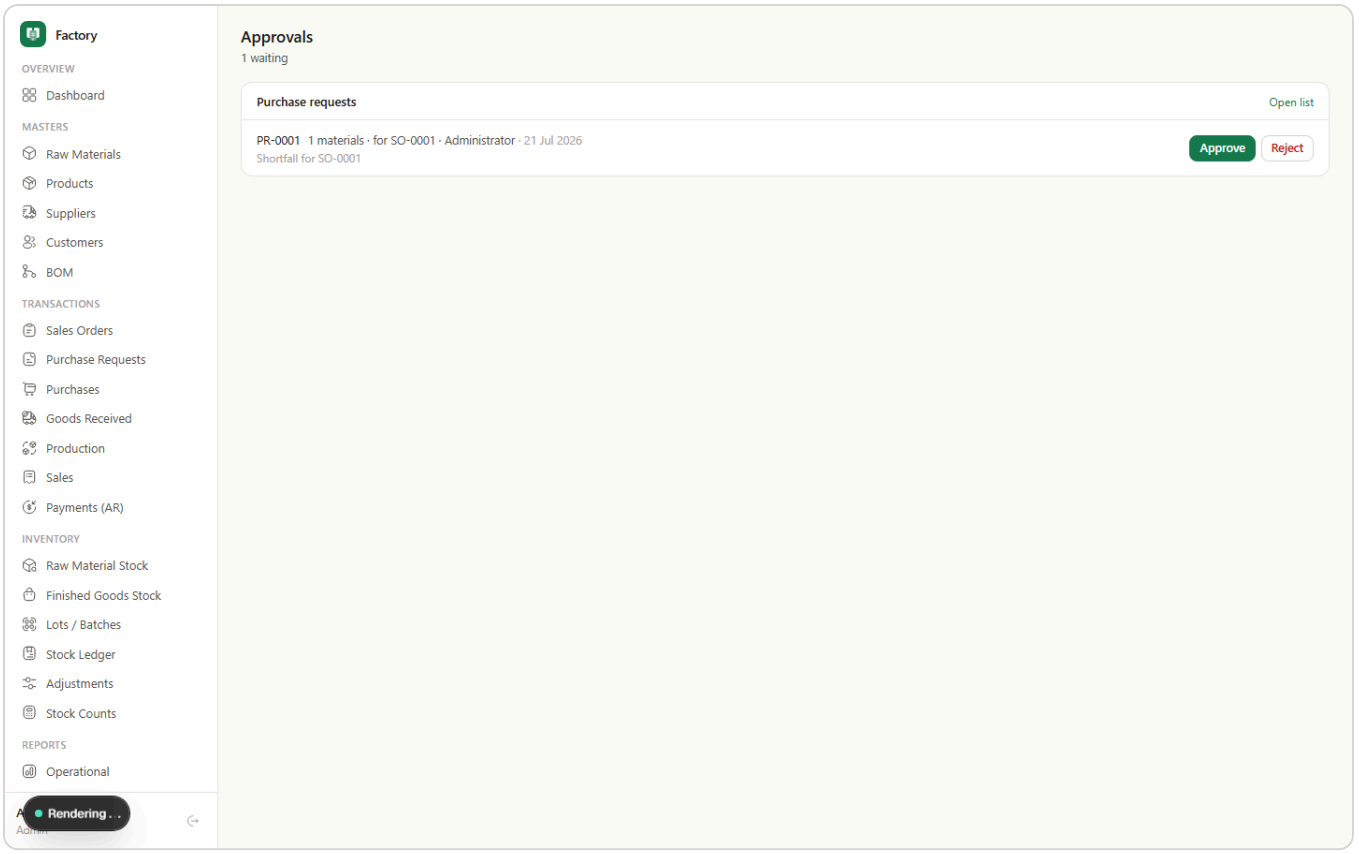
Confirming the order runs the material check. The system explodes the BOM across every line, subtracts material already reserved by other active orders, and shows exactly what is short. One button raises a purchase request for precisely the missing quantities.



Material check — PP short, one click to raise the purchase request

Approvals

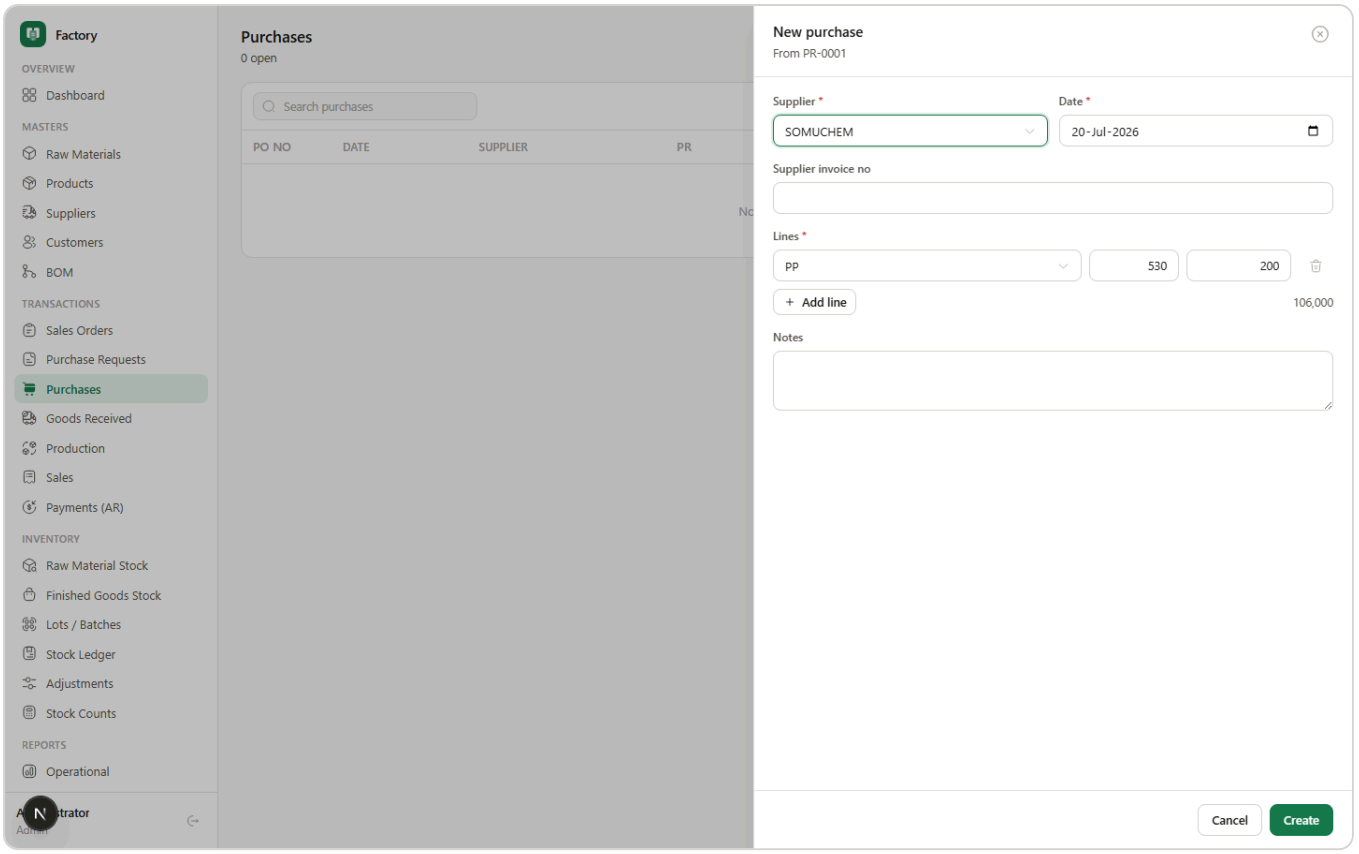
Purchase requests, stock adjustments and stock counts wait in a single approval queue. Nothing moves stock or spends money until a user with approval rights signs it off, and every decision is recorded with who and when.



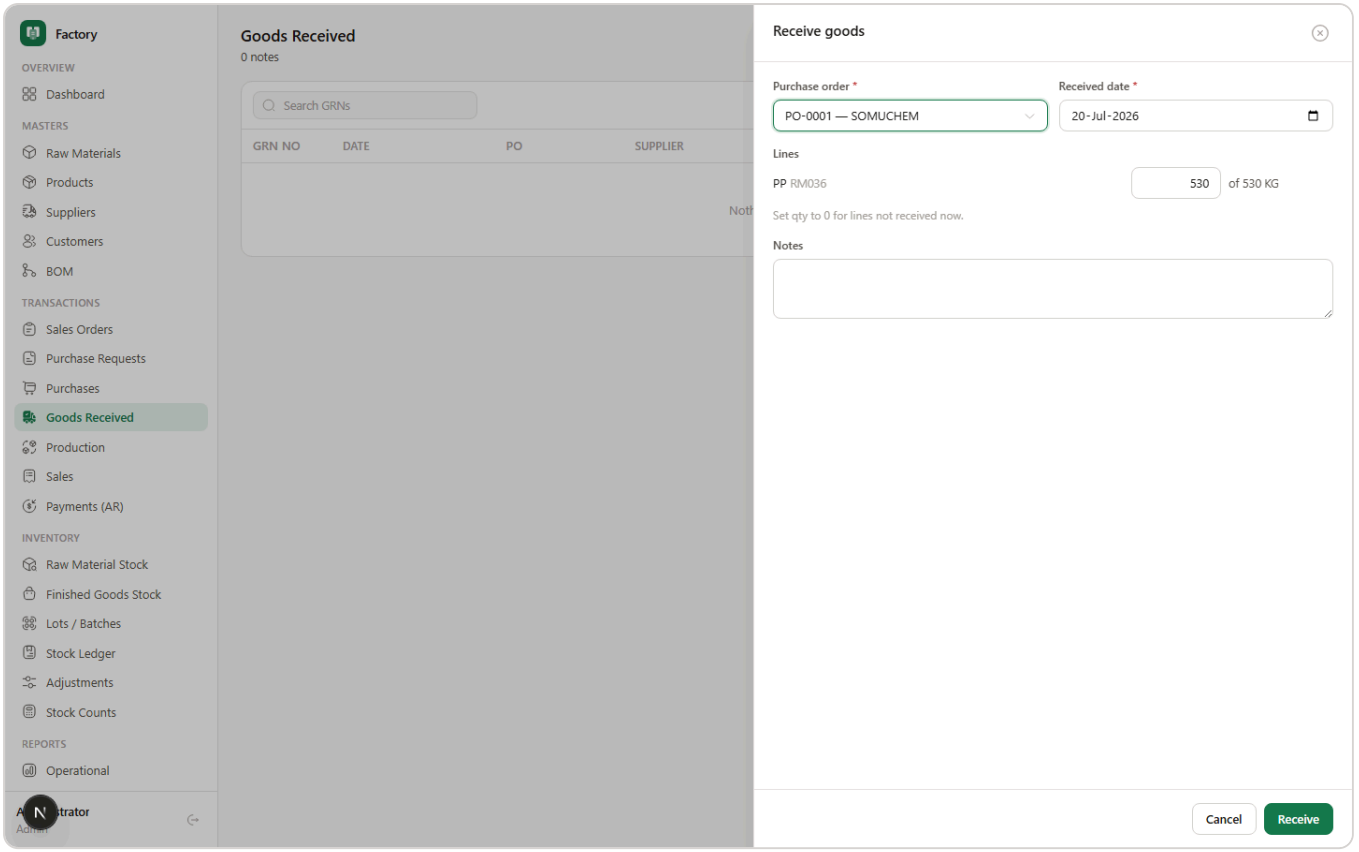
Approval queue with a pending purchase request

Purchasing and receiving

An approved request becomes a purchase order to a supplier — quantities pre-filled, rates entered at order time. Goods are received against the PO; part deliveries are supported and the order tracks `ordered → partial → received` automatically. Receiving posts stock instantly and creates a traceable lot per material.



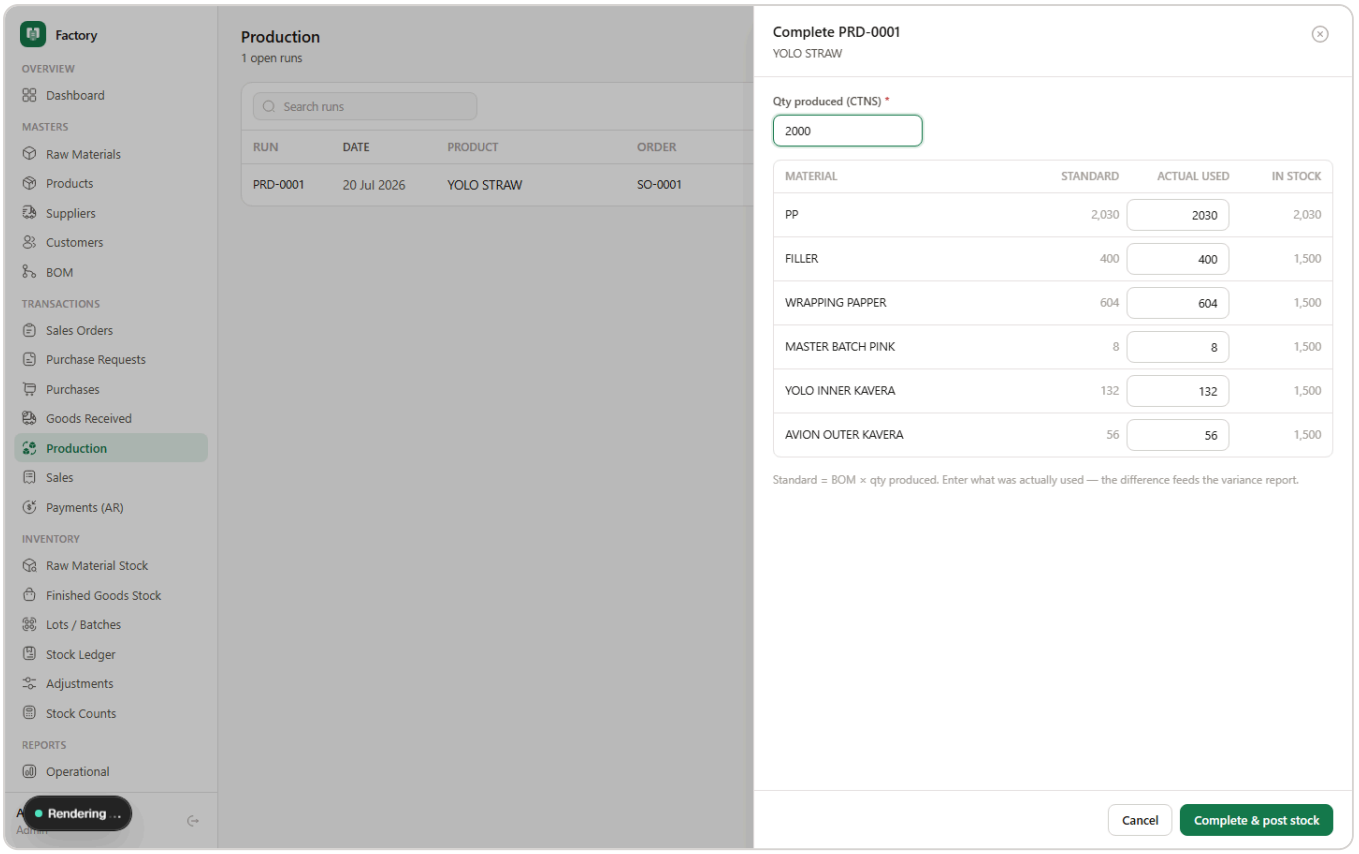
Purchase order pre-filled from the approved request



Receiving goods — outstanding quantities pre-filled

Production

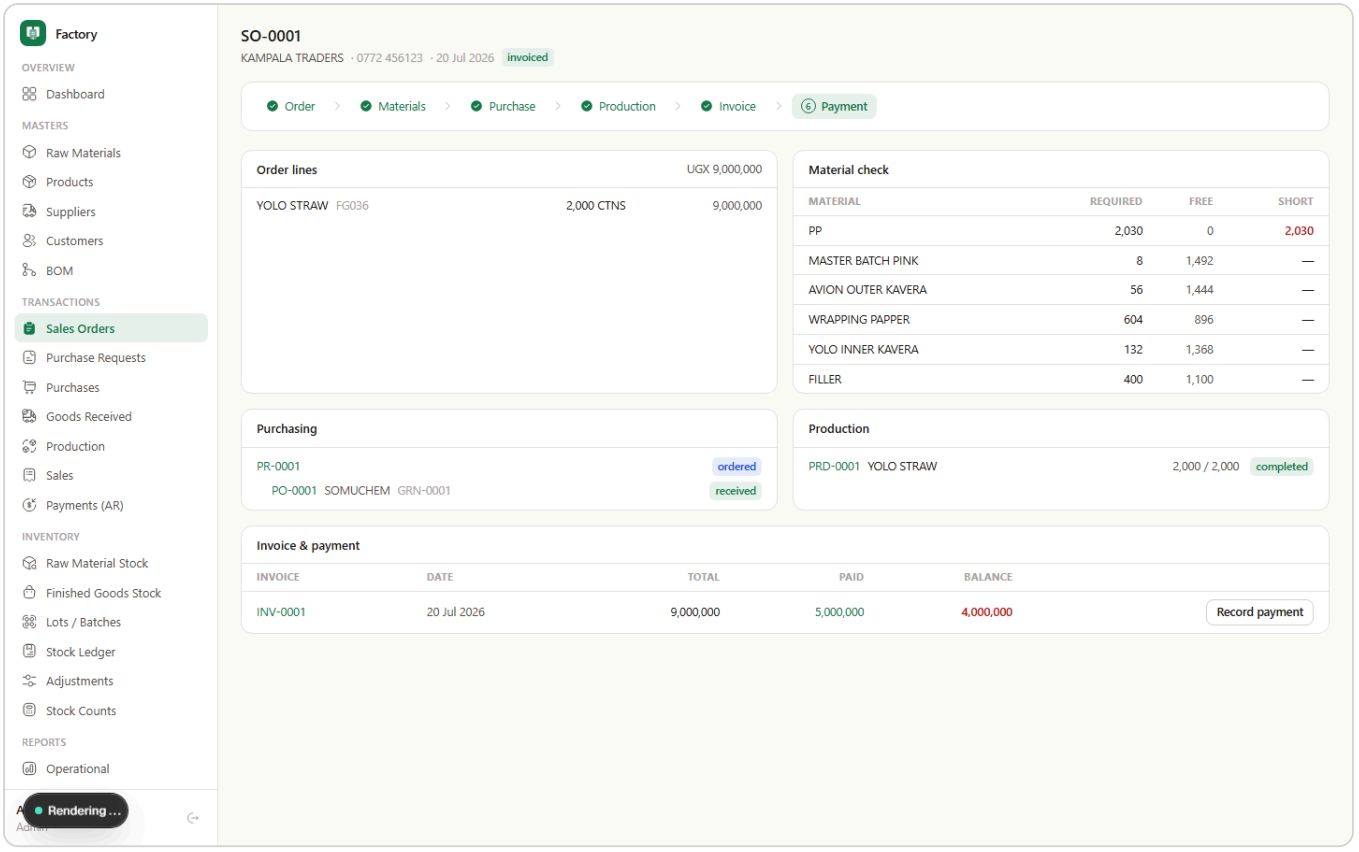
Production runs are created straight from the order (or standalone). Completing a run records the quantity produced and the actual material used — pre-filled from the BOM standard, adjusted to reality on the floor. The difference feeds the variance report, so material loss is visible per product and per material.



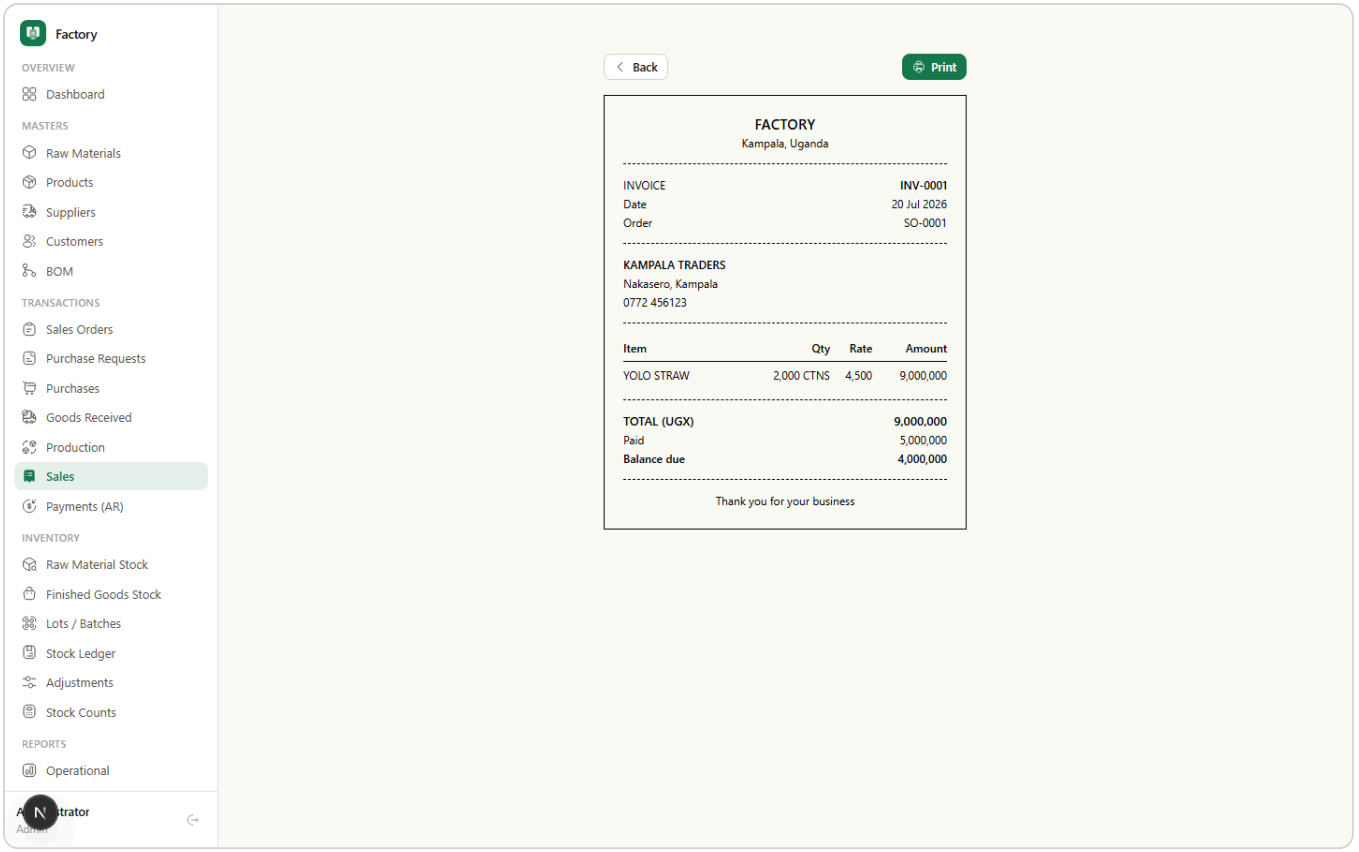
Completing a run — standard vs actual consumption, live stock check

Invoice, payment, and the finished order

When production is complete the order is invoiced in one click; finished-goods stock is deducted and the printable invoice is ready for the customer. Payments — cash, mobile money or bank — are recorded against the invoice until the balance is zero, then the order closes. The order page shows the entire chain end to end: request, PO, delivery, runs, invoice and balance.



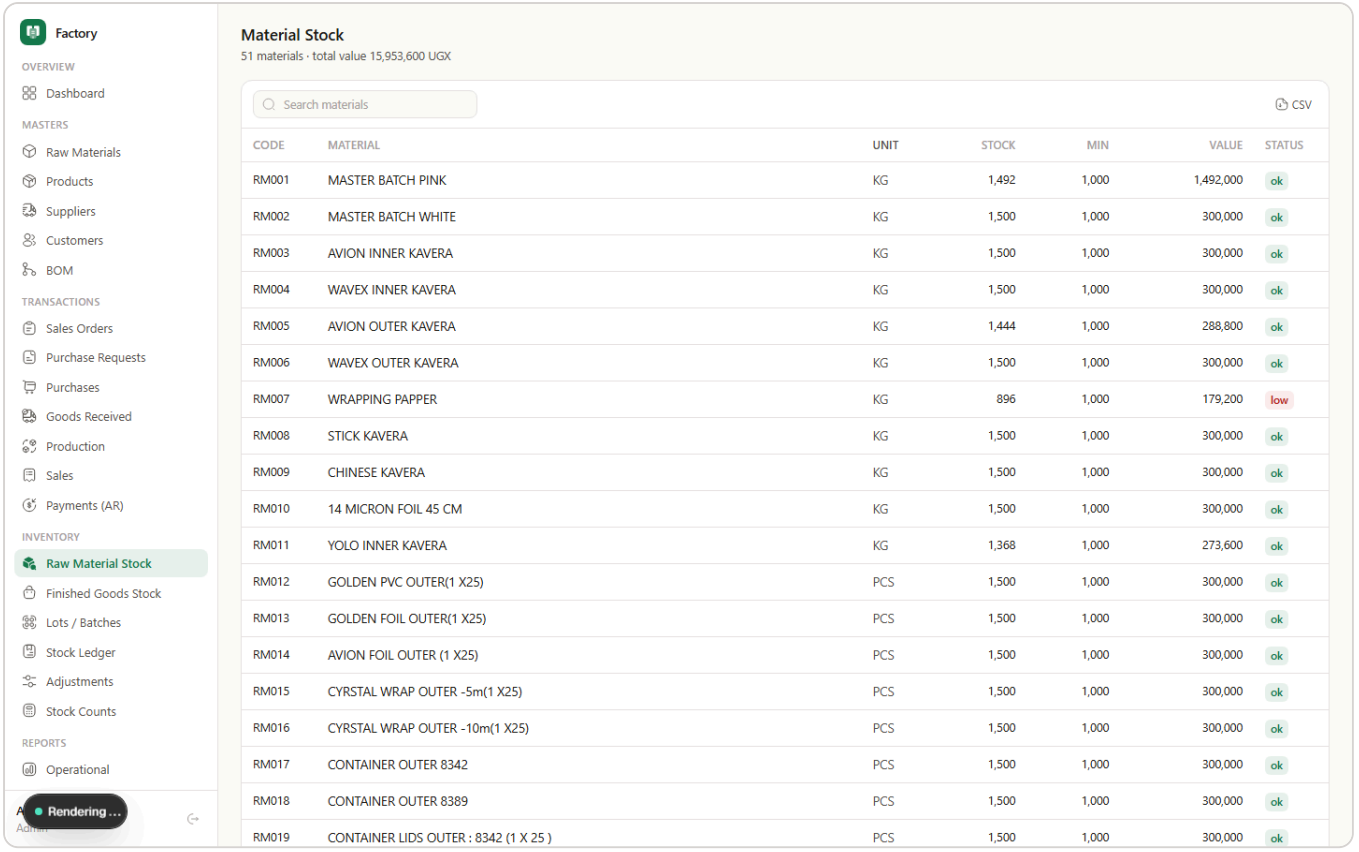
The finished order — every stage traceable on one page



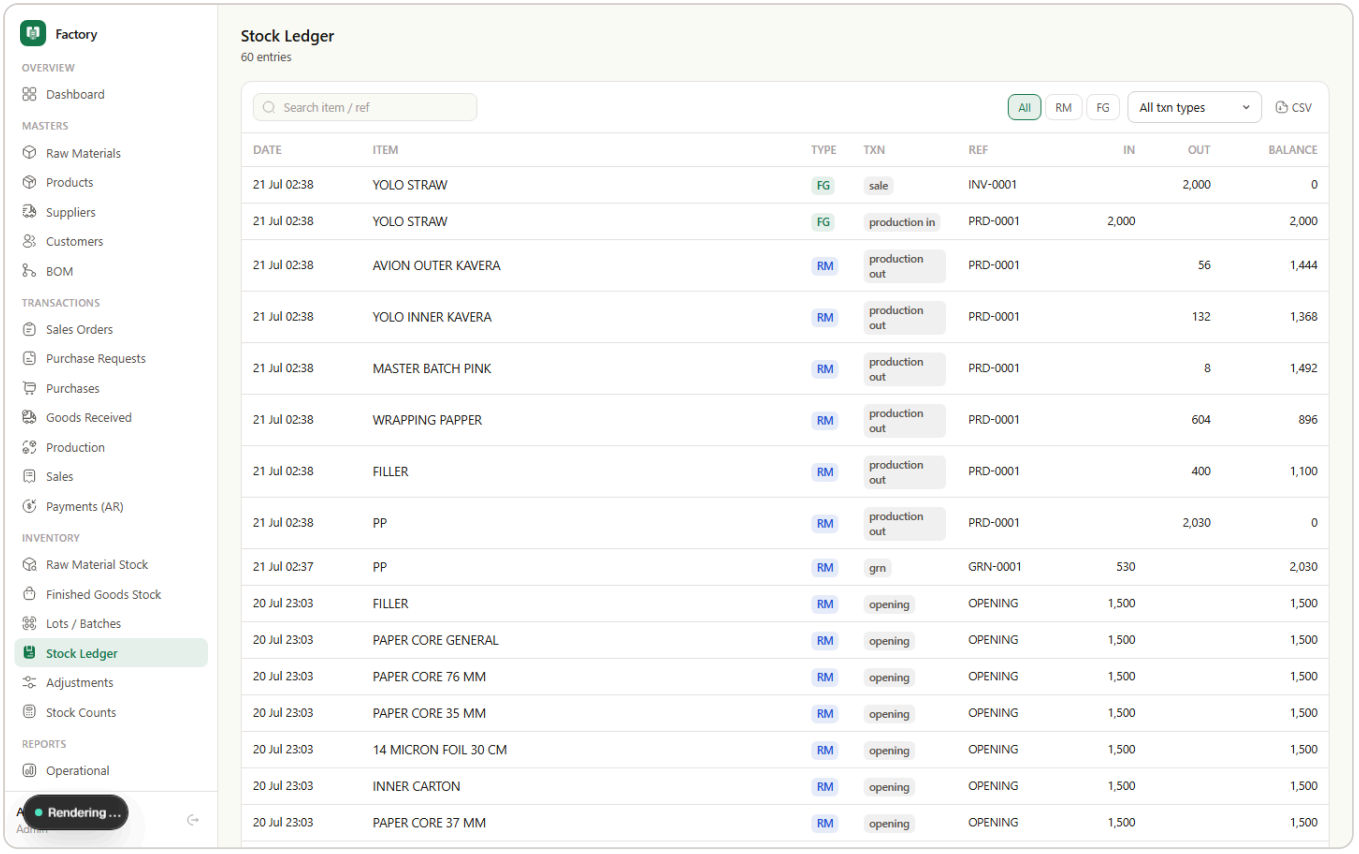
Printable invoice

Inventory control

Stock never changes silently. Every movement — opening, receipt, production in/out, sale, adjustment, count correction — is a line in the stock ledger with its document reference and running balance. Lots trace where stock came from; adjustments and physical stock counts go through approval before they correct the system.



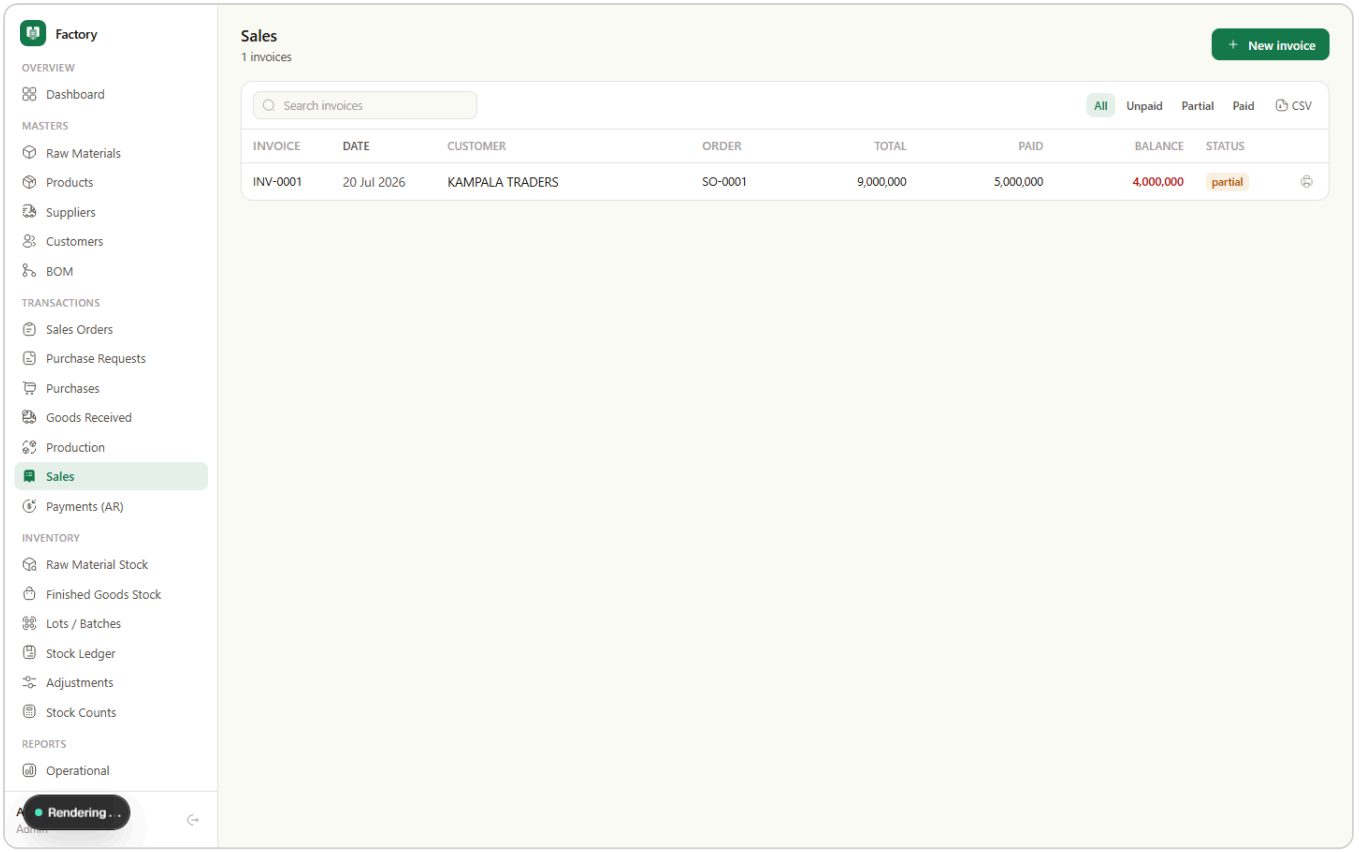
Raw material stock with values and status



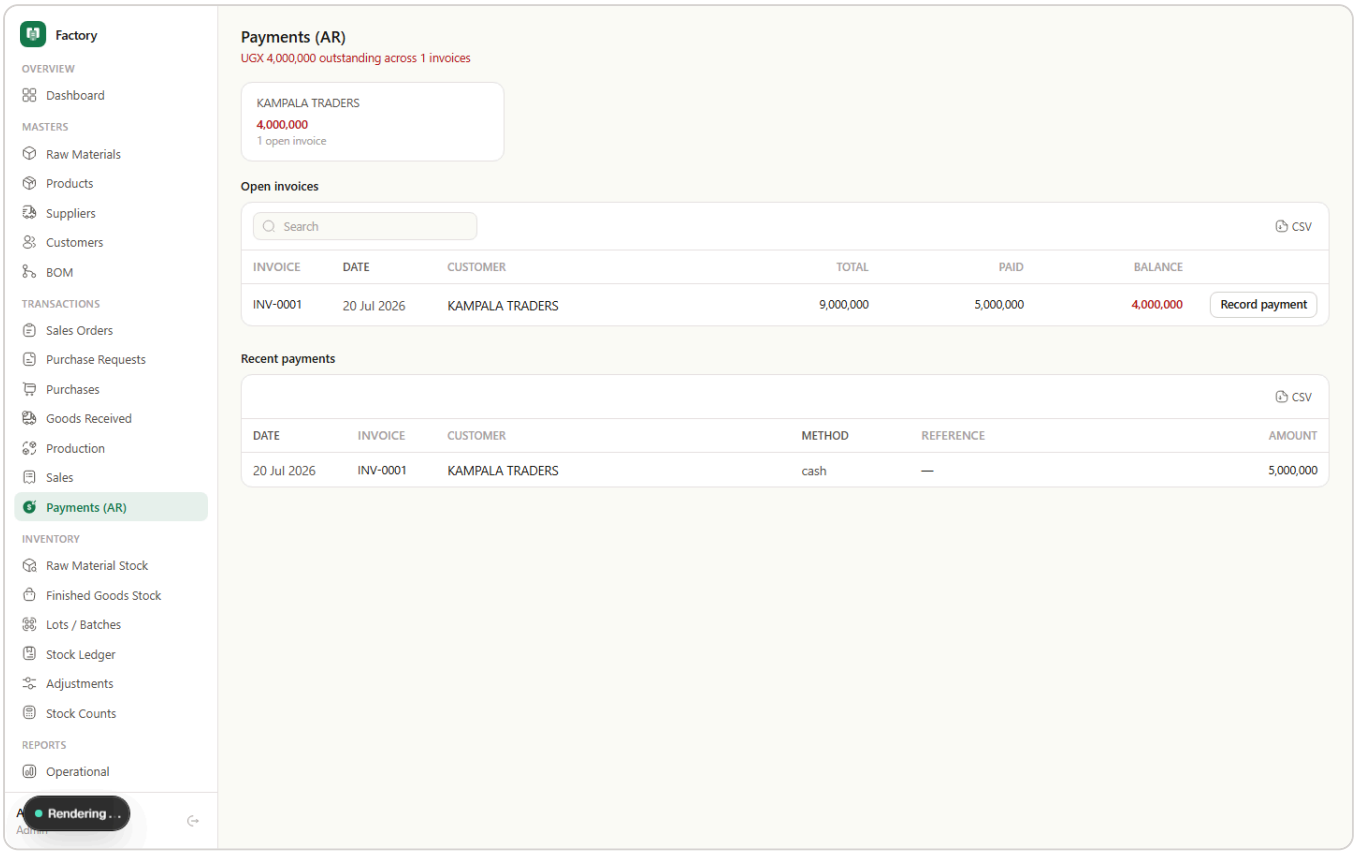
The stock ledger — every movement, its document, and the balance after

Sales and receivables

All invoices live in one register with paid and balance columns. The Payments view keeps the money conversation honest: outstanding balance per customer, open invoices, and the full payment history.

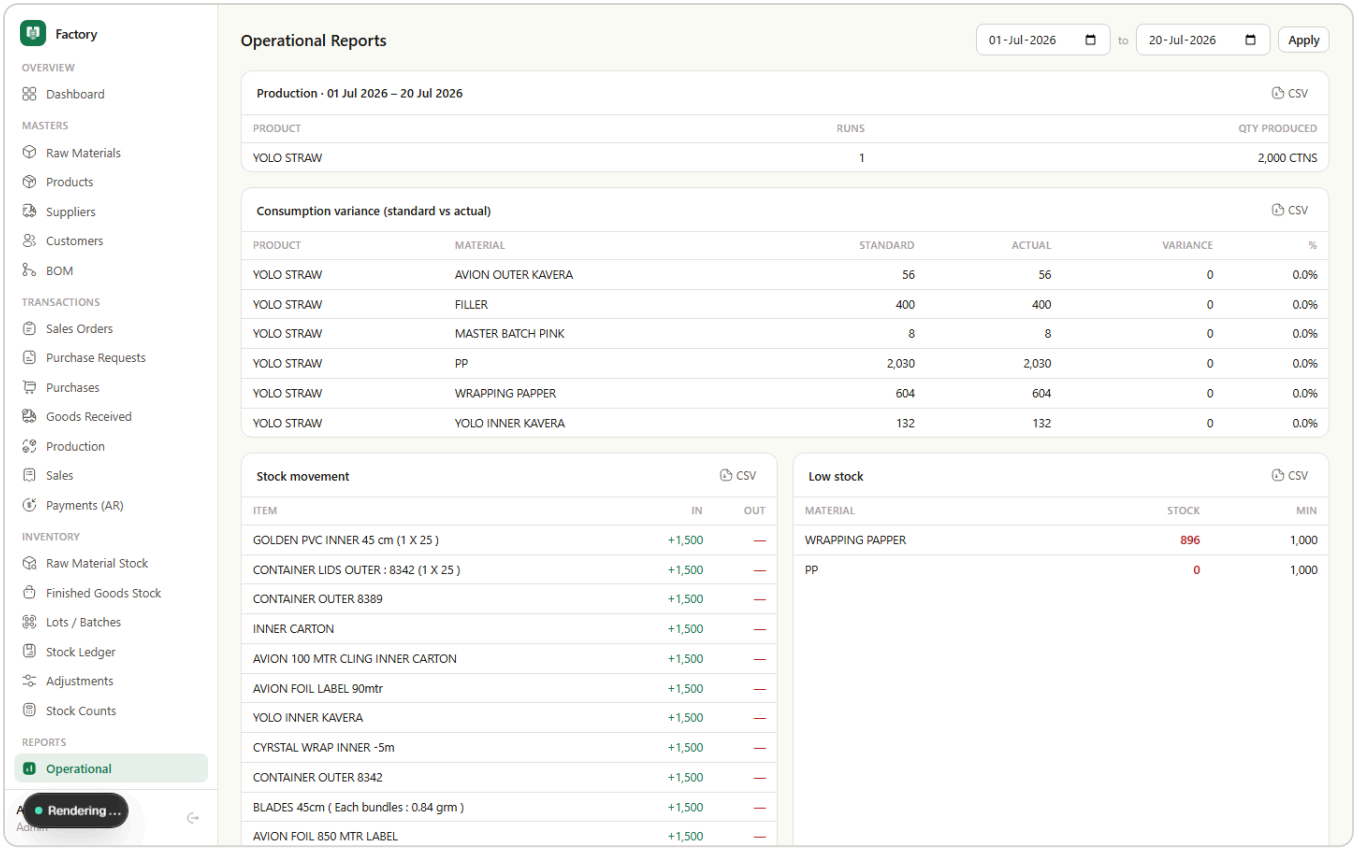


Invoice register

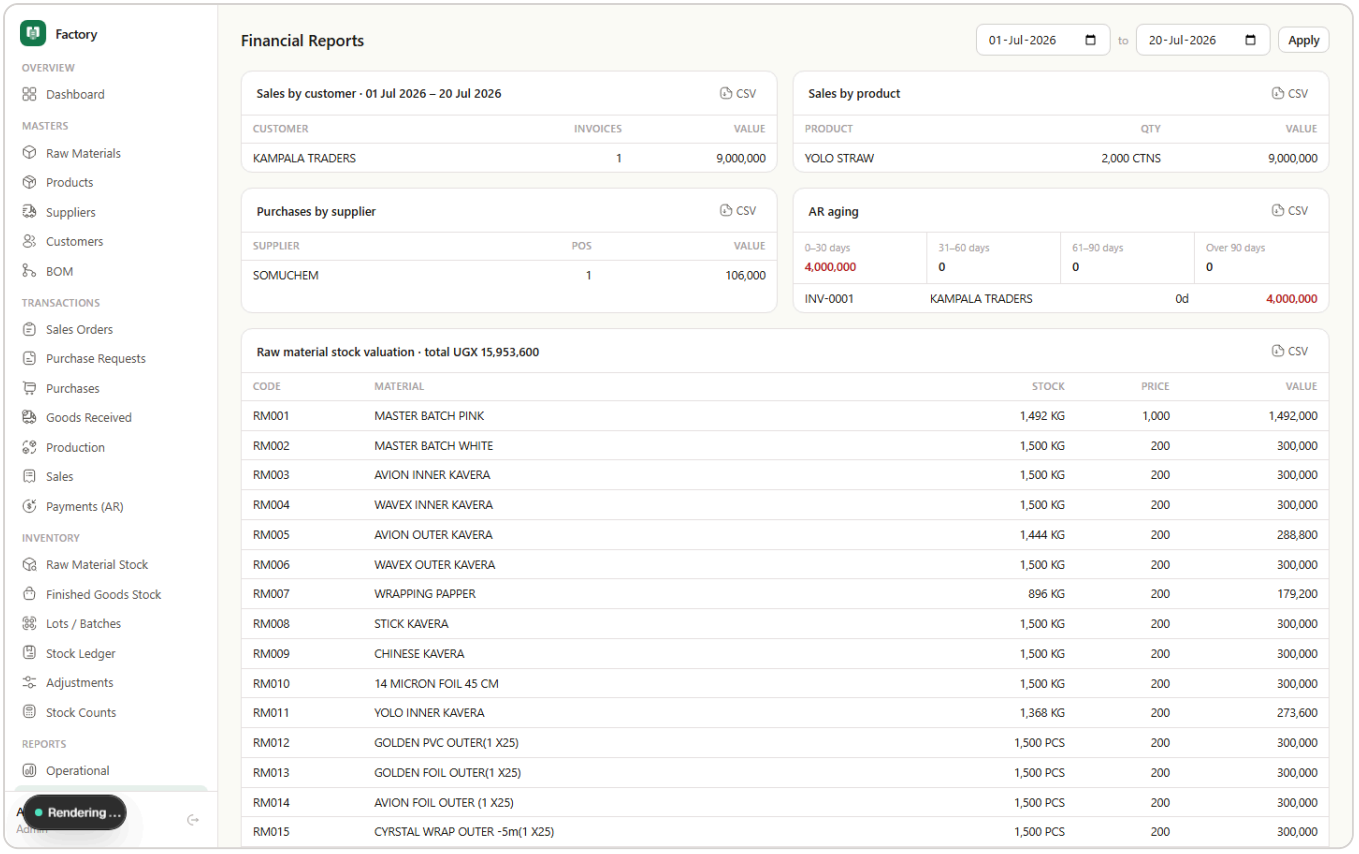


Reports

Operational reports cover production output, standard-vs-actual consumption variance, stock movement and low stock. Financial reports cover sales by customer and product, purchases by supplier, receivables aging and stock valuation. Everything filters by date range and exports to CSV — as do all the main tables in the system.



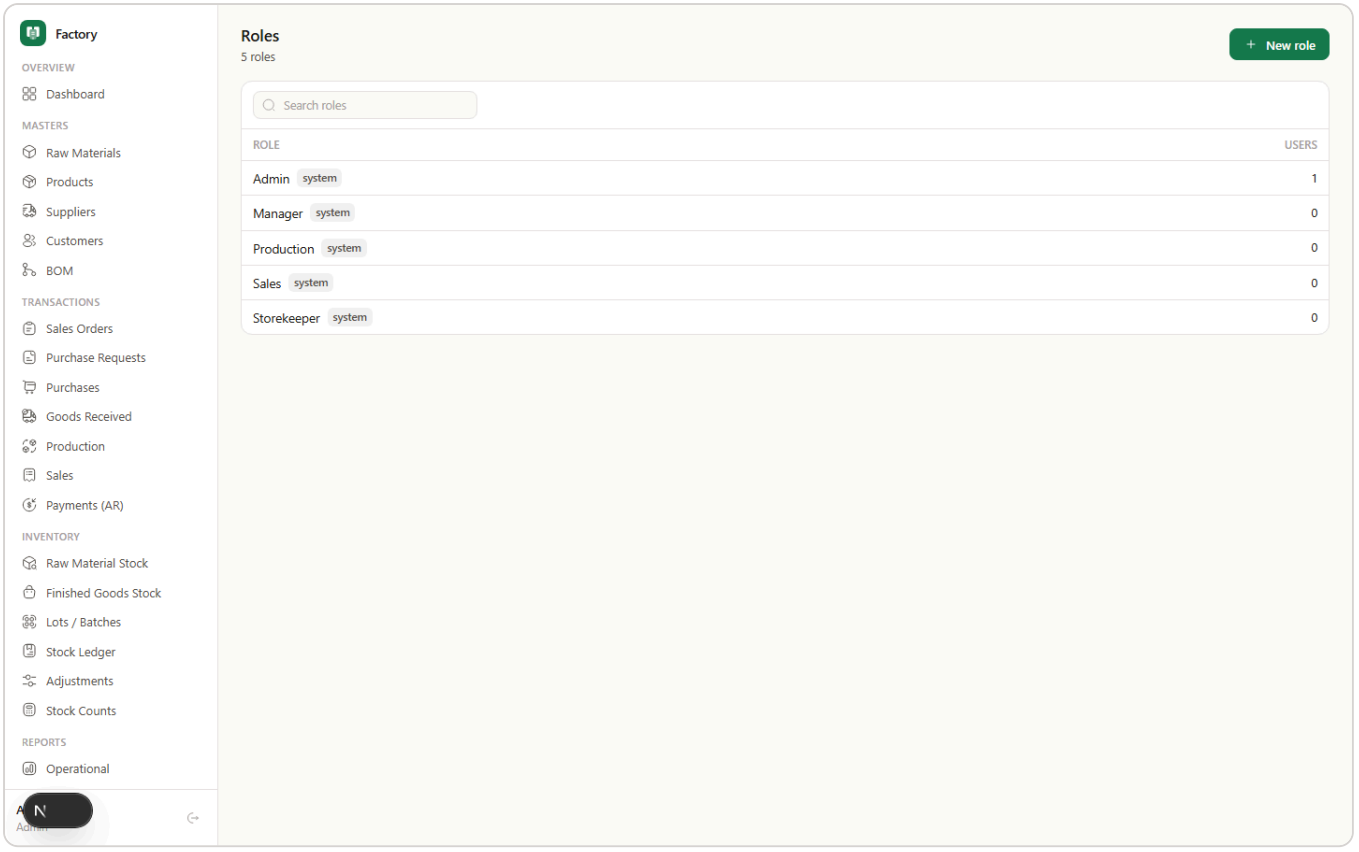
Production and variance reporting



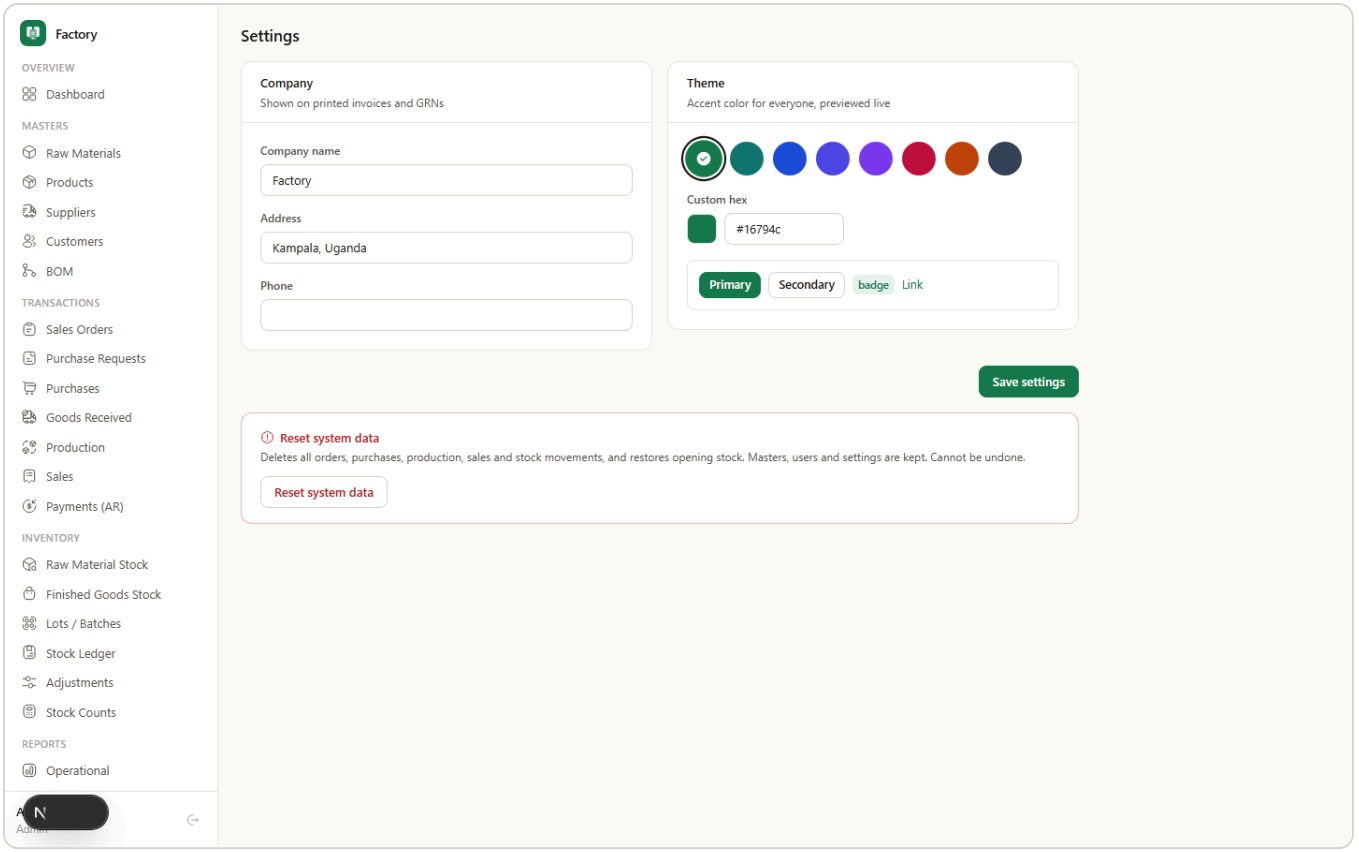
Sales, purchases, aging and valuation

Administration

Administration covers user accounts, a role–permission matrix (view / create / edit / approve per area), a full audit log of every action, and company settings including the accent colour of the interface. A security-key-protected reset can clear all transactions and restore opening stock — masters, users and the audit trail stay intact.



Roles and permissions matrix



Company settings, theme, and the protected reset

Technical summary

Application	Next.js (React) web application — works on any modern browser, no installation
Database	PostgreSQL (Neon cloud) — every stock movement in an append-only ledger
Realtime	Changes made by one user appear on all other screens within a second
Security	Individual accounts, role-based permissions, server-side checks, full audit log
Documents	Automatic numbering — SO, PR, PO, GRN, PRD, INV, ADJ, SC series
Data	37 products, 51 raw materials and complete BOM loaded from the existing workbook
Verification	Automated end-to-end test drives all 21 core scenarios through the real interface
Hosting	Deployed on Vercel; also runs on a local machine on the factory network

The system is live, seeded, and reset to opening stock — ready for real data entry.